

# Arduino Fundamentals - Assignment 1

## Reports and Answers

This document contains comprehensive answers and code solutions for all the questions across the five tasks in Assignment 1.

### TASK 1: Introduction to serial monitor.

1. **What is the difference between int and float? Give one example of when you would use each.**

An int (integer) data type stores whole numbers without fractional parts (decimals), whereas a float (floating-point number) stores numbers that include decimals.

- *Example for int:* Counting the number of iterations or tracking an age (e.g., int myAge = 21;).
- *Example for float:* Measuring precision values such as sensor voltages or temperature readings (e.g., float temperature = 25.5;).

2. **What happens if you forget to write Serial.begin(9600) in setup()? Try it and explain what you observe.**

If Serial.begin(9600); is omitted, the hardware serial interface of the Arduino remains uninitialized. As a result, the Serial Monitor will remain completely blank and show no output, because no baud rate or communication line has been established between the microcontroller and the computer.

3. **Why does the output appear only once and not over and over?**

The printing logic is placed inside the setup() function, which by design runs exactly one time when the Arduino powers on or resets. The loop() function, which executes repeatedly, was intentionally left empty for this task.

### TASK 2: Using Variables

1. **Why do we put pinMode(ledPin, OUTPUT); inside setup() and not inside loop()?**

Configuring a pin's mode is an initialization step that only needs to be performed once. Placing it in setup() is efficient and ensures proper setup before operations start, whereas putting it in loop() would redundantly reconfigure the pin thousands of times per second, wasting processing cycles.

2. **What would happen if we removed both delay() calls? Predict first, then try it and describe what you actually see.**

*Prediction:* The microcontroller executes code so rapidly that the LED will switch on and

off millions of times a second, making it appear continuously lit but faint.

*Actual Observation:* The LED stays constantly turned on at a lower or dim brightness level because the human eye cannot perceive a flashing frequency that fast (known as persistence of vision).

3. **If you wanted the LED to be ON for 3 seconds and OFF for half a second, which values would you change and to what?**

You would modify the global timing variables at the top of the program to the following values:

```
int onTime = 3000; // 3 seconds in milliseconds
int offTime = 500; // 0.5 seconds in milliseconds
```

## TASK 3: Making Functions

1. **In your own words, explain what a parameter is. (Hint: waitTime in our example.)**

A parameter acts as a placeholder variable inside a function declaration that allows external values to be passed into the function when it is called. It acts as an input that customizes the internal behavior of that function execution block.

2. **Why is it useful to put repeating code into a function instead of writing it three times in loop()?**

Encapsulating repetitive instructions reduces code duplication, rendering programs modular, compact, and considerably easier to debug or maintain. If the internal behavior needs an alteration, it only requires changing once inside the function implementation block rather than modifying multiple places.

3. **Could you write blinkOnce without taking any parameter at all? If yes, how would you make it blink at different speeds without parameters?**

Yes, it is possible. To achieve varying intervals without a parameter, you would have to rely on global variables that are reassigned or updated dynamically right before calling the function inside the loop block.

## TASK 4: Count Down with a while Loop

1. **What is the role of the line counter = counter - 1;? What would happen without it?**

This expression decrements the iteration tracker value on every single cycle to advance the evaluation closer toward breaking. Without it, the loop evaluation constraint (counter > 0) stays perpetually true, resulting in an unresolvable infinite loop locking up the Arduino.

2. **Rewrite the countdown to start from 7 instead of 5. (Paste just the changed lines in your report.)**

The modified lines would look like this:

```
int counter = 7; // Modified starting value
```

3. **In your own words, what is the difference between setup() and a while loop? They both run code - what makes them different?**

The setup() framework is a foundational function managed by the main runtime wrapper to run precisely one time at startup. A while construct is an inner programming loop structure that evaluates a boolean conditional check expression continuously, repeating a small sub-block inside either setup or loop until that condition turns false.

## TASK 5: Putting It All Together - Smart Countdown

1. **Paste your final, working code.**

```
int startValue = 5;
int ledPin = 13;

void flashLED(int times) {
    int i = 0;
    while (i < times) {
        digitalWrite(ledPin, HIGH);
        delay(200);
        digitalWrite(ledPin, LOW);
        delay(200);
        i = i + 1;
    }
}

void setup() {
    pinMode(ledPin, OUTPUT);
    Serial.begin(9600);

    Serial.println("=== Smart Countdown Starting ===");
    int count = startValue;

    while (count > 0) {
        Serial.print("Count: ");
        Serial.println(count);
        flashLED(count);
        delay(1000);
        count = count - 1;
    }
}
```

```

    Serial.println("=== Countdown Complete ===");
}

void loop() {
    // Empty loop as requested
}

```

2. **Describe one bug or mistake you ran into while writing this program and how you fixed it.**

*Example response:* A common mistake was utilizing the global count tracking variable inside the inner flashLED logic instead of declaring a separate isolated counter (like i). This caused the loop inside setup to drop to zero instantly during the first flash block. Creating an independent local tracker variable inside the function scope resolved it entirely.

3. **If you wanted the program to count DOWN by 2 each step (5, 3, 1) instead of by 1, what would you change?**

You would modify the decrementation code at the base of the while loop inside setup() from `count = count - 1;` to:

```
count = count - 2;
```

4. **If you wanted the LED to also stay ON for a final 5 seconds after the countdown completes, where in the code would you add this and what would the code look like?**

You would append the following code blocks immediately after the while loop finishes inside setup(), just before or after printing the complete completion string:

```

digitalWrite(ledPin, HIGH); // Turn LED on
delay(5000);                // Wait for 5 seconds
digitalWrite(ledPin, LOW);  // Turn LED off

```